

SHAH NAWAZ ALI ASGHAR

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EDUCATION

National University of Sciences and Technology (NUST), Islamabad, Pakistan Oct 2022 – May 2026
B.E. in Computer Software Engineering | **CGPA: 3.13 / 4.00** (Graduated)
Thesis: MasSecEval – AI-Powered SAST Platform Combining Static Analysis and OpenTelemetry Execution Tracing
Relevant Coursework: Linear Algebra, Calculus & Analytical Geometry, Probability & Random Processes, Data Structures & Algorithms, Analysis of Algorithms, Operating Systems, Software Architecture

RESEARCH INTERESTS

Safe and Constrained Reinforcement Learning, Lagrangian methods for Constrained MDPs, Multi-Agent Systems & LLM Agents, Computer Vision for Physical Actuation, Trustworthy AI, Sequential Decision-Making under Constraints.

PUBLICATIONS & PREPRINTS

- **Shah Nawaz Ali Asghar**, "Curriculum Constraint Tightening in Lagrangian PPO: A Controlled Empirical Study of λ Stability and Generalization in Discrete Navigation," preprint, 2026. [PDF] [Code]
Empirical study of dual-variable dynamics under curriculum constraint schedules. Identified a key trade-off: curriculum stabilizes λ (0.38–0.42) but yields a weaker final safety signal and lower zero-shot success on unseen maps (56.3% vs. 75.3% for fixed-target baseline).
- Aun Shahid, M. Usman Baig, **Shah Nawaz Ali Asghar**, Umar Hanif, "Browser-Based Gesture Control for Multi-Device Smart Home Appliances Using IoT," Technical Report, Military College of Signals, NUST, 2025/2026. [PDF]
YOLOv11 detection + MediaPipe Handpose (21 landmarks) + spatial selection for camera-based multi-device control via ESP32/Blynk.

PROFESSIONAL EXPERIENCE

- Artificial Intelligence Engineer + Backend** Aug 2025 – Present
AI Connect Hub (Managed by Growlatics) – Remote (Sydney, Australia)
 - Architected and deployed production multi-agent systems (LangGraph/LangChain) for voice, marketing, health, meeting, and administrative workflows.
 - Integrated Azure AI Foundry for cloud-native inferencing; optimized system latency and multi-user reliability.
- Backend + AI Intern** Oct 2023 – Feb 2024
Incept AI – Onsite
 - Built prompt-chaining pipelines and sandbox testing workflows for a prompt-to-code SaaS product.
 - Structured complex user intents into modular code generation tasks to reduce hallucination risk.

PROJECTS

- MasSecEval – AI-Powered SAST Platform** (Final Year Thesis)
 - Engineered a multi-agent pipeline combining static source code scanning with OpenTelemetry dynamic execution tracing.
 - Integrated Qdrant vector store and Gemini RAG for OWASP Top-10 aligned vulnerability reasoning and ranked findings.
- Enterprise Knowledge + SQL Agent**
 - Built a unified LangGraph system for document RAG (FAISS/S3) and natural-language SQL over SQLite/MySQL, supporting PDF, DOCX, CSV, XLSX, and image ingestion.
- Admin AI & Meeting Assistant**
 - Developed an end-to-end multi-agent SaaS (LangGraph, FastAPI, Microsoft Graph) for form automation, scheduling, and Teams meeting join/transcript/summary workflows.
- Additional Machine Learning Projects**
 - Aspect-Based Sentiment Analysis: Fine-tuned RoBERTa on ~8,374 aspect-labeled posts (Accuracy: 0.8386, F1: 0.8381).
 - Reinforcement Learning Benchmarks: Implemented Dueling DQN on LunarLander and comparative PPO vs. DQN on CartPole.
 - Computer Vision & Security: Detectron2 Mask R-CNN & Mask2Former segmentation; Network Intrusion Detection with SHAP explainability; MedicalQA LoRA fine-tuning (DeepSeek-7B).

TECHNICAL SKILLS

- **Reinforcement Learning:** Lagrangian PPO, Constrained MDPs, DQN / Dueling DQN, PPO, Curriculum Methods, PyTorch
- **Multi-Agent & LLMs:** LangGraph, LangChain, RAG (FAISS, Qdrant), Prompt Engineering, Agentic Workflows
- **Computer Vision & IoT:** YOLOv11, MediaPipe / Handpose, Detectron2, Mask2Former, ESP32, Edge Actuation
- **Languages & Tools:** Python, C++, JavaScript/TypeScript, FastAPI, Docker, Azure, SQL, Git

ADDITIONAL INFORMATION

- Medium of Instruction: English (Official Certificate available from NUST Registrar).
- Target Intake: Spring 2027 Graduate Program (KAIST).
- Research Papers: github.com/shahnawaz05/Research-Papers